

torch.optim.Optimizer.state_dict#

[https://pytorch.org/docs/stable/generated/torch.optim.Optimizer.state_dict.h](https://pytorch.org/docs/stable/generated/torch.optim.Optimizer.state_dict.html)

Description:

Return the state of the optimizer as a dict. It contains two entries: differs between optimizer classes, but some common characteristics hold. For example, state is saved per parameter, and the parameter itself is NOT saved. state is a Dictionary mapping parameter ids to a Dict with state corresponding to each parameter. parameter group is a Dict. Each parameter group contains metadata specific to the optimizer, such as learning rate and weight decay, as well as a List of parameter IDs of the parameters in the group. If a param group was initialized with `named_parameters()` the names content will also be saved in the state dict. NOTE: The parameter IDs may look like indices but they are just IDs associating state with param_group. When loading from a state_dict, the optimizer will zip the param_group params (int IDs) and the optimizer param_groups (actual nn.Parameter s) in order to match state WITHOUT additional verification.

Function Signature

```
Optimizer.state_dict()[source]#
```

state: a Dict holding current optimization state. Its content

param_groups: a List containing all parameter groups where each

Return type

Returns:

dict[str, Any]

Code Examples

```
{
  'state': {
    0: {'momentum_buffer': tensor(...), ...},
    1: {'momentum_buffer': tensor(...), ...},
    2: {'momentum_buffer': tensor(...), ...},
    3: {'momentum_buffer': tensor(...), ...}
  },
  'param_groups': [
    {
      'lr': 0.01,
      'weight_decay': 0,
      ...
      'params': [0]
      'param_names' ['param0'] (optional)
    },
    {
      'lr': 0.001,
      'weight_decay': 0.5,
      ...
      'params': [1, 2, 3]
      'param_names': ['param1', 'layer.weight',
'layer.bias'] (optional)
    }
  ]
}
```

Detailed Documentation

Documentation

Return the state of the optimizer as a dict.

It contains two entries:

differs between optimizer classes, but some common characteristics hold. For example, state is saved per parameter, and the parameter itself is NOT saved. state is a Dictionary mapping parameter ids to a Dict with state corresponding to each parameter.

parameter group is a Dict. Each parameter group contains metadata specific to the optimizer, such as learning rate and weight decay, as well as a List of parameter IDs of the parameters in the group. If a param group was initialized with `named_parameters()` the names content will also be saved in the state dict.

NOTE: The parameter IDs may look like indices but they are just IDs associating state with `param_group`. When loading from a `state_dict`, the optimizer will zip the `param_group` params (int IDs) and the optimizer `param_groups` (actual `nn.Parameter s`) in order to match state WITHOUT additional verification.

A returned state dict might look something like: